## WATER PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

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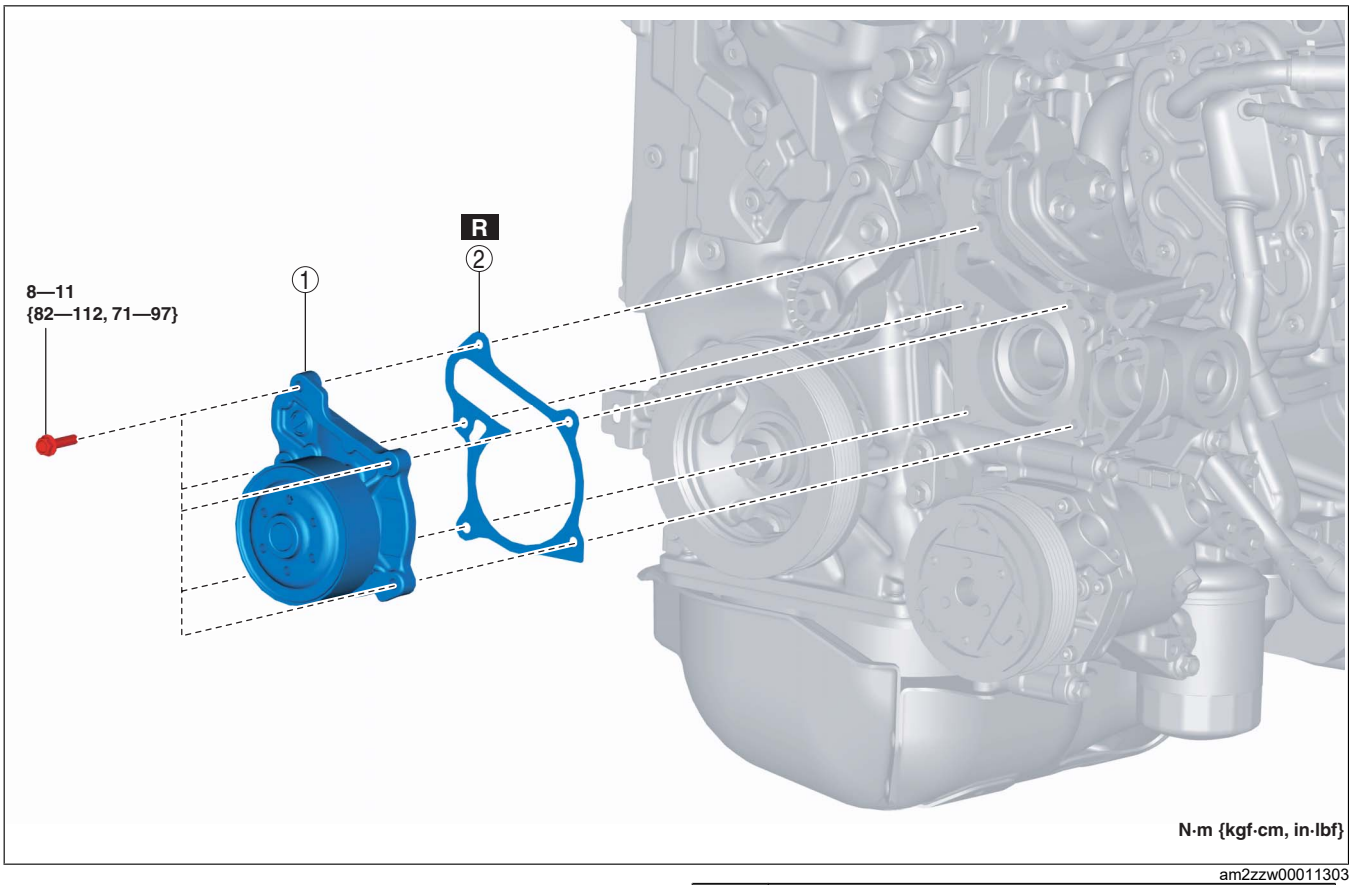
**Warning**

- **If the cooling system cap or the water-cooled charge air cooler reserve tank cap is removed when the engine is hot, steam and boiling water may be injected, causing severe burns or severe injury. Perform the removal of the cooling system cap and water-cooled charge air cooler reserve tank cap when the engine is cool.**
- **Turn off the engine and wait until it is cool. Even then, be very careful when removing the cap. Wrap a thick cloth around it and slowly turn it counterclockwise to the first stop. Step back while the pressure escapes.**
- **When you are sure all the pressure is gone, press down on the cap using the cloth, turn it, and remove it.**

**Caution**

- **Be careful not to damage the exterior circumference surface of the water pump pulley when using tools, otherwise it will cause wear, breakage, abnormal noise of the drive belt, damage to the pulley, and rust.**
- **If coolant leakage from the water pump is considered, refer to [Determination Criteria for Water Pump Coolant Leakage] to determine. (See Determination Criteria for Water Pump Coolant Leakage.)**

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
3. Remove the front splash shield (RH). (See SPLASH SHIELD REMOVAL/INSTALLATION.)
4. Remove the drive belt. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Disconnect the service plug. (with i-ELOOP) (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)
7. Remove the generator. (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].)
8. Drain the engine coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
9. Remove in the order indicated in the table.
10. Install in the reverse order of removal.
11. Refill the engine coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
12. Inspect for engine coolant leakage. (See ENGINE COOLANT LEAKAGE INSPECTION [SKYACTIV-D 1.5].)




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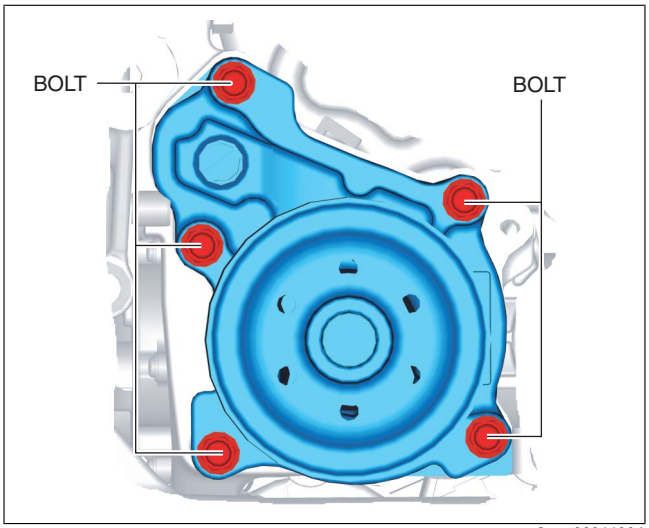
| | |
|---|---|
| 1 | Water pump component<br>(See Water Pump Component Removal Note.)<br>(See Water Pump Component Installation Note.) |
| 2 | Gasket |

**Water Pump Component Removal Note**

**Caution**
- **Be careful not to damage the exterior circumference surface of the water pump pulley when using tools, otherwise it will cause wear, breakage, abnormal noise of the drive belt, damage to the pulley, and rust.**

1. Slightly loosen the five water pump component bolts shown in the figure and collect the draining engine coolant into a container.

**Caution**
- **Do not remove the water pump component completely at this time because the engine coolant remaining in the engine will drain out.**



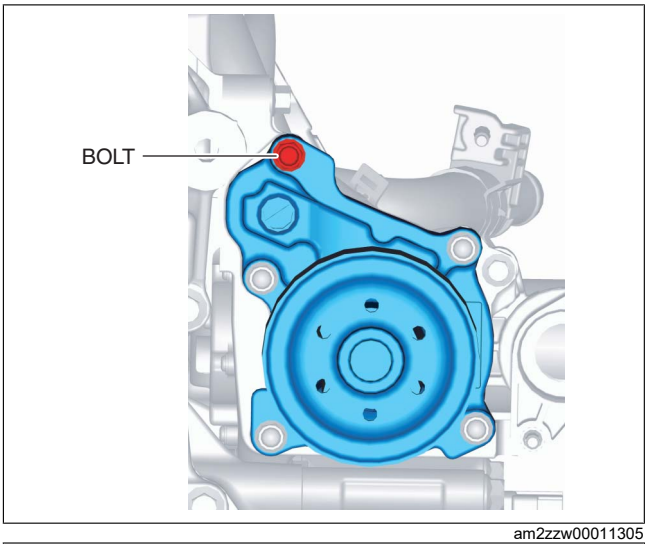

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### Water Pump Component Installation Note

**Caution**

- **Be careful not to damage the exterior circumference surface of the water pump pulley when using tools, otherwise it will cause wear, breakage, abnormal noise of the drive belt, damage to the pulley, and rust.**

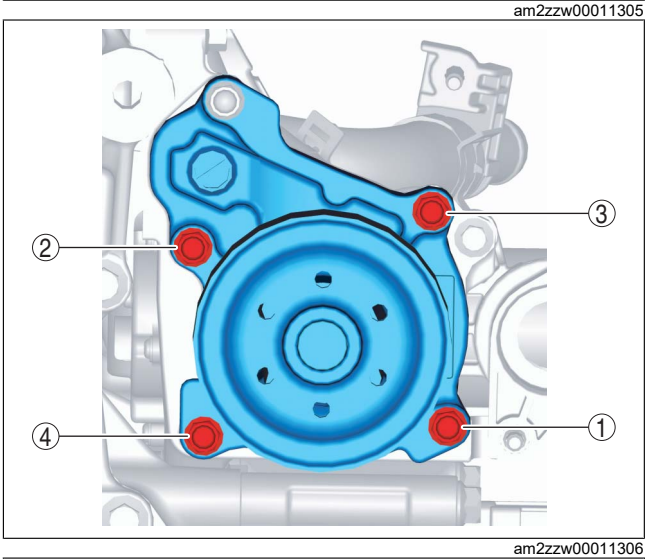
1. Install the water pump component using the following procedure:
   (1) Temporarily tighten the bolt shown in the figure.



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   (2) Tighten the bolts in the order shown in the figure.

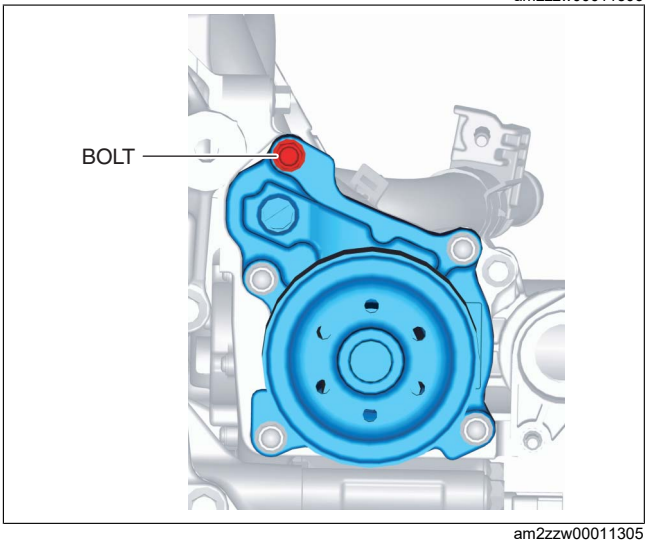
**Tightening torque**
**8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}**



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   (3) Finally, tighten the bolt shown in the figure to the specified torque.

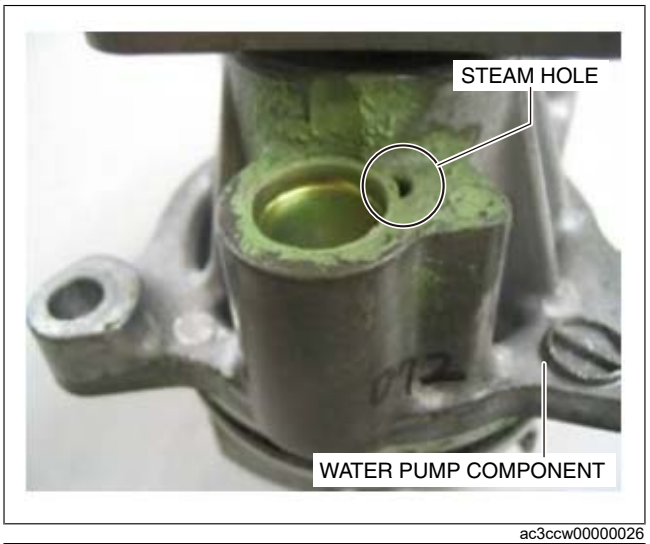
**Tightening torque**
**8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}**



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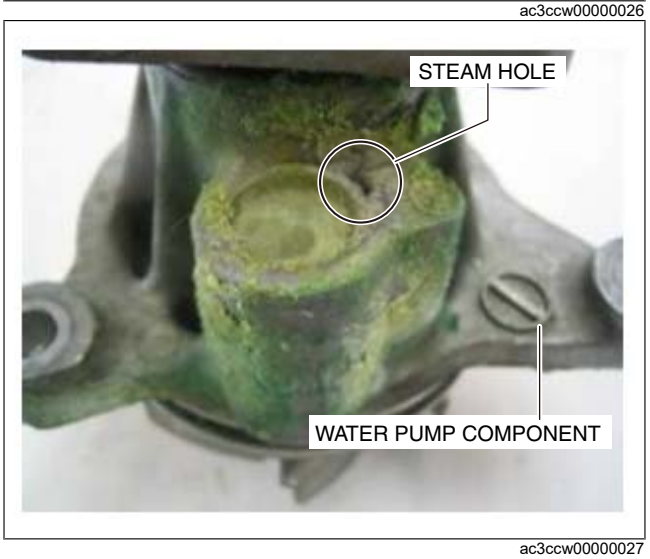
**Determination Criteria for Water Pump Coolant Leakage**

- If any coolant residue (green) around the steam hole can be verified but there is no large solid matter adhered to it as shown in the following figure, the residue is caused by a small amount of steam blowing through the steam hole, not coolant leakage. Therefore, this does not indicate a problem.



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- If a large amount of solid matter is adhered to the area around the steam hole or the area is wet by coolant as shown in the following figure, a water pump seal or gasket malfunction can be considered. Therefore, perform the coolant leakage inspection and replace the water pump or gasket. (See ENGINE COOLANT LEAKAGE INSPECTION [SKYACTIV-D 1.5].)



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